

Efrain Rocha

Director of Engineering
Distributed Systems, Data & Platform Architecture

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SUMMARY

Director of Engineering with 25+ years of experience modernizing and operating business-critical software platforms in regulated, data-intensive environments. Deep background in distributed and event-driven systems, data and cloud platforms at scale, and platform reliability, with a consistent focus on performance, maintainability, and long-term system health.

Known for taking brittle, slow, or high-risk systems and re-engineering them into resilient, service-oriented platforms that teams can evolve safely. Brings strong technical judgment, pragmatic decision-making, and a proven ability to deliver under real deadlines where failure would directly impact revenue, customers, or regulatory commitments.

TECHNOLOGY DEPTH & BREADTH

Core Strengths: Distributed & event-driven architectures, messaging systems, data platforms at scale, cloud reliability, system performance

Languages: Python, Shell, SQL, Java, C/C++; working knowledge of C#, JavaScript

Messaging & Integration: RabbitMQ (request/response & async), event-driven workflows, legacy API replacement, batch-to-message migrations

Data Platforms: PostgreSQL (90+ TB), MySQL, Oracle, MSSQL Server; partitioning, sharding, replication, retention strategies

Cloud & Infrastructure: AWS, Oracle OCI, Linux/Windows, infrastructure-as-code, CI/CD pipelines

Engineering Leadership: SDLC ownership, architectural governance, design reviews, regulated environments, delivery under SLA constraints

PROFESSIONAL EXPERIENCE

U.S. LawShield - Houston, TX

Senior Development Manager | Platform Architect

2024 – Present

Modernized a brittle, revenue-impacting backend platform by replacing a decade-old, tightly coupled PHP API layer with a service-oriented, message-driven architecture, enabling faster feature delivery, improved performance, and reduced customer-impacting incidents.

Improved response times, system reliability, and maintainability, significantly reducing the risk of breaking changes and enabling teams to evolve functionality safely.

Decoupled core systems using RabbitMQ-based request/response messaging, allowing services to scale independently and fail gracefully.

Direct business impact: enabled faster delivery of new features and refinements, reduced customer support tickets caused by slow or unresponsive systems and improved the platform's ability to meet member SLAs.

Gene by Gene - Houston, TX
Senior Development Manager | Enterprise Architecture
2015 – 2024

Reduced DNA analysis turnaround from ~30 days to ~5 days and core processing time from ~6 hours to ~20 minutes, enabling the business to keep pace with lab throughput and meet strict customer delivery SLAs.

Built and led engineering, DevOps, and database teams responsible for high-throughput genomic and laboratory platforms operating under strict regulatory and delivery constraints.

Delivered a highly scalable, distributed processing platform using service-oriented and message-driven architectures backed by a 90+ TB PostgreSQL data platform.

Led a critical 3-week delivery to adapt analysis systems after lab technology changes—work that would normally take 2–3 months—preventing missed deliveries and customer impact.

Partnered with Oracle OCI to deliver a cloud-based laboratory platform supporting time-critical, high-volume diagnostic processing.

Re-engineered legacy batch and file-based pipelines into resilient, message-driven workflows capable of sustained, large-scale execution.

Operated large, distributed compute workloads (100+ VMs continuously) to re-analyze 1.6M DNA samples, completing months-long processing safely and reliably.

Audubon Engineering - Houston, TX
Senior Manager, CAD/CAE Software Applications
2014 – 2015

Delivered and stabilized internal engineering platforms used across upstream, midstream, downstream, and pipeline operations, supporting mission-critical workflows for engineering and project teams.

Directed application architecture and integrations supporting engineering analysis, project workflows, and enterprise systems.

JPMorgan Chase & Co. - Houston, TX
Vice President, Physical Logistics Core
2010 – 2014

Led delivery of a reusable physical logistics platform supporting Natural Gas trading across the Americas and EMEA, enabling consistent, scalable operations for business-critical trading workflows.

Directed the architecture and delivery of a shared logistics framework that enabled faster expansion into new commodities and regions.

Supported business-critical trading and scheduling workflows where system failures would directly impact market operations and financial risk.

Integrated with Endur and enterprise Oracle data warehouses to support trading, logistics, and operational reporting.

Earlier Leadership Roles

Director, Product & Architecture - Codefast, Inc.

Director of Engineering & Quality Assurance - SunGard Energy

Director, Enterprise Systems Architecture - El Paso Global Networks

2001 - 2007

Led Director-level engineering organizations delivering long-lived enterprise platforms in telecom and energy, including systems at WorldCom/MCI and Verizon that remained operational for 20+ years.

Accountable for long-term system reliability, platform evolution, and delivery execution across mission-critical enterprise systems.

EDUCATION

M.S., Computer Science — Monterrey Institute of Technology (ITESM)

B.S., Computer Science Engineering — Monterrey Institute of Technology (ITESM)